

Cloud security concerns	
Category	Concern
Identity and access management	Authentication
	SSO
	Authorisation
	User directory and federation services
Data security	Data confidentiality
	Data integrity
	Availability
	Back-ups and archives
	Key management
Network security	Security for data in transit
	Perimeter security
	Network security threats (DoS, man-in-the-middle, packet sniffing)
OS and virtualisation security	Virtualisation/Hypervisor threats
	Host OS
	OS hardening
	Anti-virus

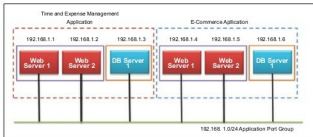


Figure 3: Application deployment on a private cloud

provided by specific private cloud products.

- Ensure complete isolation between Application 1 and Application 2
- Isolate one Web server from another in each application
- Allow HTTP (80)/HTTPS (443) traffic to Web servers
- Allow Web server to database server communication on a specific DB port
- Block all other traffic
- Security policies to be completely transparent to IP address changes and network renumbering
- Dynamic rules to be applied to new virtual machines automatically
- Default 'Deny All' policy in case of firewall rules

The solution

This solution gives organisations deep visibility into network communications between virtual machines, and eliminates the hardware and policy sprawl associated with using traditional appliances to secure virtualised

Best practices to ensure security in a cloud environment	
• Design for failure • Geo-distributed data centre • Back-up of data on the cloud • Full back-up • Incremental back-up • Layer-wise security • Web layer • App layer • Data layer	• Default 'Deny all' policy • Logical data centre • Geo-distributed app architecture • Software-based data replication • Data durability • Use of Memcache • Monitoring services • Distribute static and dynamic content

applications. In addition to that, it optimises hardware resource utilisation while maintaining strong levels of security.

Security offerings from proprietary and open source initiatives enable a hypervisor-based application-aware firewall to dynamically define and enforce policies for virtual machines that are supporting business critical applications. Customers can define a security group, for instance, to protect a group of virtual machines that might hold credit card data that needs to be PCI compliant, or another group of virtual machines that hold patient health records and needs to be HIPAA compliant.

Firewall rules to accomplish the requirements of this use case must be defined. When a new virtual machine joins the container or security group, the rules set-up will be applied automatically and there will be no need to define new rules.

Advantages

- Eliminates the complexity and cost of using security groups to segregate traffic. An application firewall with deep visibility helps to filter traffic between virtual machines.
- Monitors 'flows' of traffic, independent from firewall rules.
- Reports are based on applications, sessions, bytes and other details, allowing administrators to define and refine their policies based on actual traffic patterns.

Identity management

In an organisation, multiple business units deploy their applications in the private cloud. In addition to that, customers also use private cloud resources. The challenge is to provide secure access to all consumers of a private cloud. The security and identity infrastructure in the private cloud is an important management platform component. Private clouds can be based on VMware or OpenStack products.

The solution

Multiple directory services are used in organisations such as OpenLDAP, which is a free, open source